MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

1/ TYPE DOD-006

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Any
Weight: 2 pounds (907.2 grams) nominal

ABSOLUTE RATINGS: See note 4

Grid pulsed:

Parameter:	Ef	Eb	Ec2	Ec1	ib	Ib	Pg1	Pg2	Pp	tk	Cooling
Unit:	V	kVdc	kVdc	Vdc	a	Adc	W	W	kW	---	---
	(Note 1)				(Note 2)						(Note 3)
Maximum:	6.0	5.0	1.2	-300	18	0.2	30	50	1.5	300	---
Minimum:	5.0	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS: (Note 5)	5.5	2.5	0.5	Vary	---	0.6	---	---	---	300	Note 3

GENERAL:

Qualification - Required

1/ Replaces RCA type 4629
NSN 5960-00-824-9391

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
Ⓐ	Qualification inspection								
---	High-temperature operation	16	TA = +85° C	---	---	---	---	---	---
---	Low-temperature operation	16	TA = -55° C	---	---	---	---	---	---
1042	Shock, specified pulse	12, 14	Test condition A; no voltages applied	---	---	---	---	---	---
	Quality conformance inspection, part 1								
1366	Interelement leakage resistance, cold	6	Supply voltage (source) = 1,000 Vdc; Rs = 20,000 to 410,000 ohms; vary to give full scale deflection with external leads shorted together	0.65	II	Rg1-k Rg1-g2 Rg2-p	5 10 20	--- --- ---	Meg Meg Meg
1266	Total grid current	7	t = 300 sec	0.65	II	Ic1	----	-300	μAdc
1261	Electrode voltage (grid)			0.65	II	Ec1	-9.5	-26.5	Vdc
1256	Electrode current (screen)			0.65	II	Ic2	-35	+7.0	mAdc
---	Cutoff leakage	-	Eb = 5.0 kVdc; Ec2 = 400 Vdc; Ec1 = -120 Vdc	0.65	II	Ib	---	25	μAdc
1301	Heater current			0.65	II	If	16.3	18.2	A
1231	Pulsing emission (sinusoid)	8, 13	eb = ec2 = ec1; etd/ik = 120 a; tp = 2 μs; prr = 60; t = 60 (max)	0.65	II	etd	---	1,000	v
---	Pulsed operation	11	Ebb = 4,000 Vdc; Ecc2 = 400 Vdc; Ecc1 = -120 Vdc; egk = 120 v; edriver = +240 v; Ef = 5.1 V; tp = 200 ±30 ns; prr = 1,000 ±10%	0.65 0.65	II II	ib ic2	8.0 ---	--- 1.4	a a
	Quality conformance inspection, part 2								
---	Lead pull	9	No voltages applied	4.0	S4	---	---	---	---
1331	Direct-interelectrode capacitance	10		---	---	Cg1-k Cg1-g2 Cg1-p Cg2-k Cg2-p Cp-k	37 46 --- --- 14.6 ---	46 62 0.17 1.40 17.8 0.017	pF pF pF pF pF pF
	Quality conformance inspection, part 3								
---	Life test	15	Pulsed operation; t = 500 hours	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life-test end point:								
---	Pulsed operation			---	---	ib	8.0	---	a
				---	---	lc2	---	1.4	a
1032	Vibration, mechanical	12, 15	Eb = 450 Vdc; Ec2 = 300 Vdc; Ec1/Ib = 10 mAdc; Rℓ = 2,000 ohms Axis of application (see figure 3) Frequency range = 10 to 50 to 10 Hz Vibration level = 0.080-inch total displacement Time of sweep = 3 to 15 minutes Time at resonance = 60 seconds	---	---	Ep	---	500	mVac

NOTES:

1. Heater surge current shall be limited to 55 amperes maximum.
2. For pulse duration (tp) and pulse repetition rate (prr) other than specified, see figure 4.
3. Airflow shall be supplied to limit the seals and anode core temperature to 250°C maximum.
4. Ratings apply for a maximum "on" time of 10 μs in any 1,000 μs interval.
5. All tests shall be performed without the leads attached with the exception of the pulsing emission (sinusoid) and lead-pull test.
6. Tube to be tested at room temperature. The resistance between the electrodes shall be measured with a Biddle 1,000 volt megger, or equivalent. Resistance to be measured in both directions.
7. Airflow shall be supplied to limit the anode core temperature to 250°C (min) to 275°C (max). All other seals shall be limited to 250°C.
8. This test is to be the first test performed at the conclusion of the holding period.
9. The lead-pull test shall be performed by applying a weight of 19 pounds to each lead in the direction of the longitudinal axis. A tube having one or more defective leads shall be considered a defective tube. Electrical rejects may be used
10. All unused electrodes grounded. When measuring - Cg1-p, Cp-k: Use external flat-metal shield 8-inches in diameter and having a center hole approximately 3-inches in diameter with spring fingers that connect to grid 2 terminal. Shield located in plane of grid 2 terminal, perpendicular to the tube axis.

    Cg2-k, Cp-k: Use external flat-metal shield 8-inches in diameter and having a center hole approximately 2.375-inches in diameter with spring fingers that connect to grid 1 terminal. Shield located in plane of grid 1 terminal, perpendicular to the tube axis.
11. The grid shall be pulsed positively to obtain the specified peak grid voltage shown on figure 2.
12. There shall be no shorts or opens after this test. Tubes shall meet the limits for the following tests: Total grid current, electrode voltage (grid), and electrode current (screen). This test may be performed prior to the attachment of tube leads.

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NOTES: -Continued

13. tp = 2 μs minimum and not greater than 10 μs at the base line of the pulse.

14. The tube shall be subjected to 18-impact shocks (see figure 3 for orientation of axes). Three shocks in each of the mutually perpendicular axes in the following sequence: Y+, Y-, X+, X-, Z+, and Z-. This is not a destructive test.

(A) 15. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

(A) 16. With static general test conditions applied.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

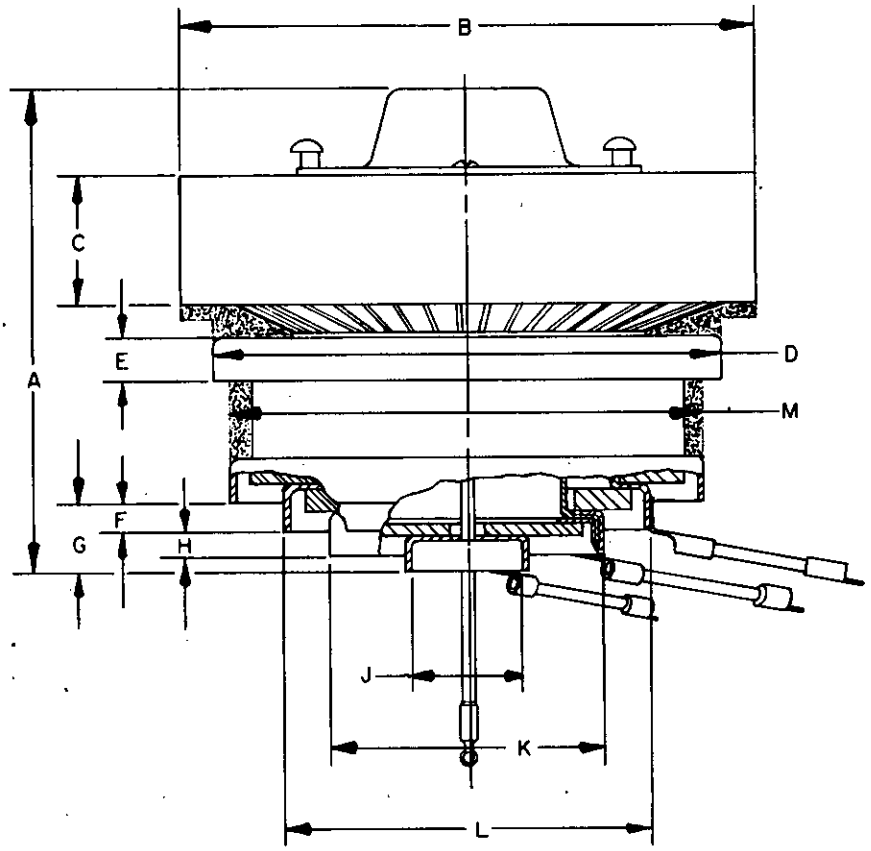
User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3040)

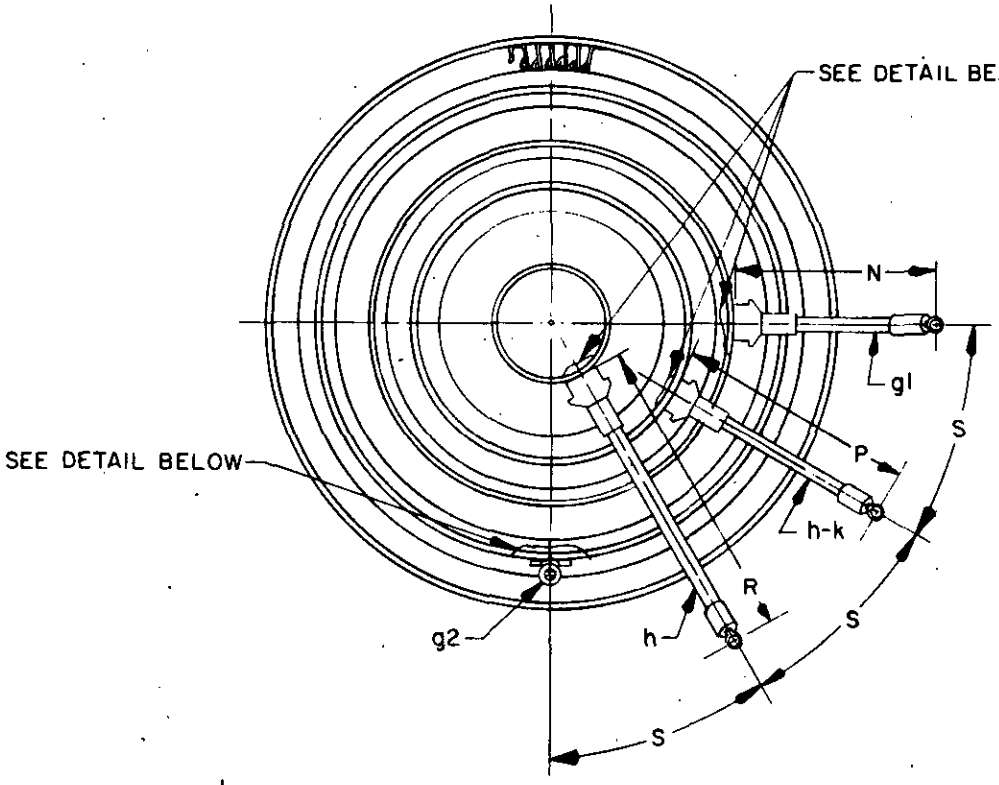
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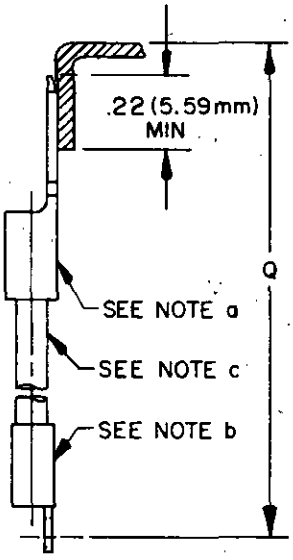


KEEP ALL STIPPLED REGIONS CLEAR.
DO NOT ALLOW CONTACTS OR CIRCUIT COMPONENTS TO PROTRUDE INTO THESE ANNULAR VOLUMES.

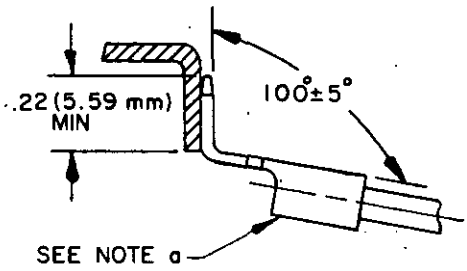
FIGURE 1. Outline drawing of electron tube type DOD-006.

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DETAIL OF g2 LEAD ASSEMBLY



DETAIL OF h, h-k, and gl LUGS

FIGURE 1. Outline of drawing of electron tube type D0D-006 - Continued.

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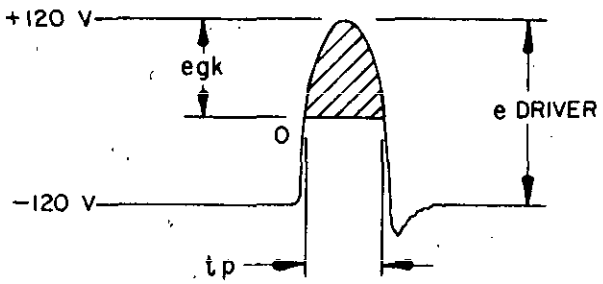
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 2	
A	3.140(79.76)	3.340(84.84)
B	3.690(93.73)	3.750(95.25)
N	1.300(33.02)	1.700(43.18)
P	1.500(38.10)	1.900(48.26)
Q	1.600(40.64)	2.000(50.80)
R	2.000(50.80)	2.400(60.96)
S	27°	33°
	Quality conformance inspection, part 3 (see note d)	
C	.850(21.59)	
D	3.210(81.53)	
E	.220 (5.59)	
F	.175 (4.45)	.225 (5.72)
G	.430(10.92)	.490(12.45)
H	.115 (2.92)	
J		.725(18.42)
K	1.710(43.43)	
L	2.307(58.60)	
M	3.010(76.45)	

Ⓐ

NOTES:
a. ETC Inc. type MCT-1 lug, or equivalent (crimp).
b. AMP Inc. Cat. No. 32440 ring, or equivalent (MS25036-6) (crimped).
c. All wire leads shall be No. 14 AWG, 41 strand tin plated copper wire (uninsulated).
d. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type D0D-006 - Continued.

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GRID DRIVE VOLTAGE

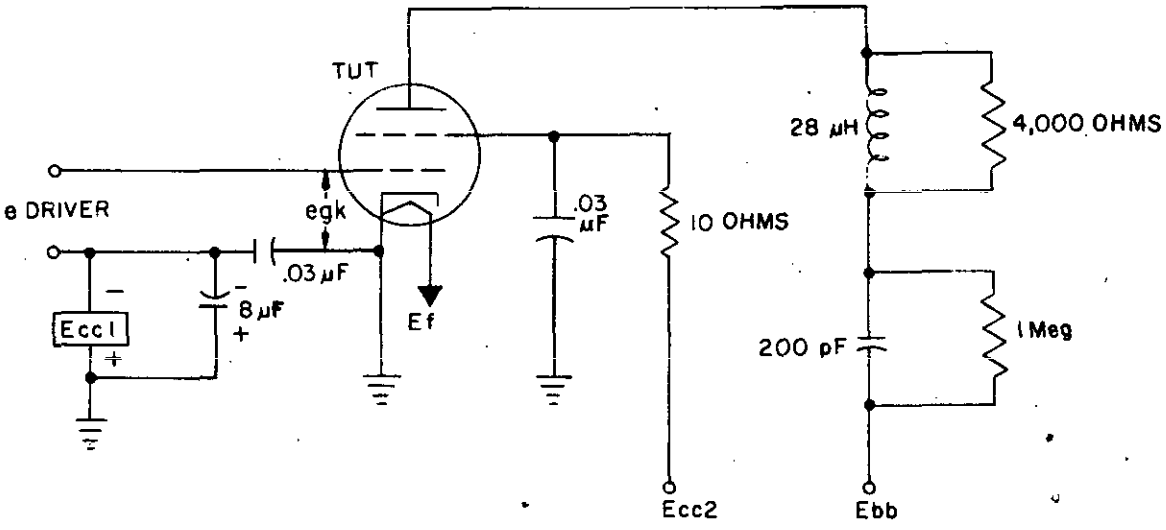
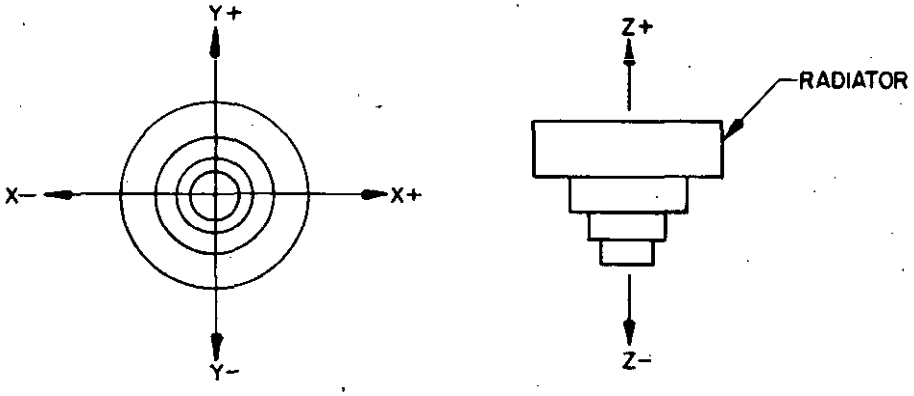


FIGURE 2. Test circuit for pulsed operation.



TOP VIEW SIDE VIEW

FIGURE 3. Orientation of axes.

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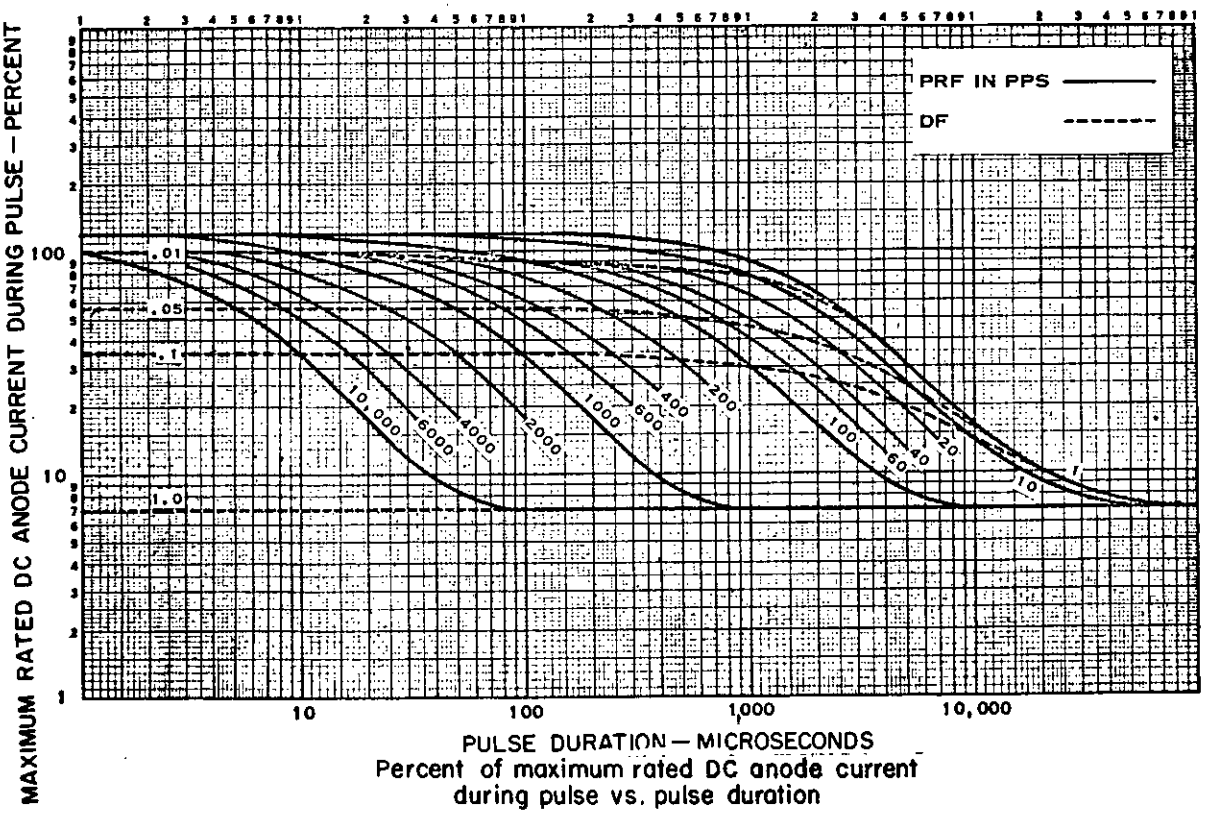


FIGURE 4. Constant PRF and DF characteristics chart.

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